Task 1C - Submission

⚠ Plagiarism check

Every submitted file is checked with plagiarism-detection software. A plagiarised submission may lead to disqualification of the team.

How It Is Graded

Task 1C carries **50 marks**. Grading happens in two places, and only the second one counts towards your score:

Where	On which clips	What it is for
On your machine, with the evaluation tool	the **public** clips you were given	Your own feedback, and the result file you upload
On the portal, after you submit	a **private** set of clips you have never seen	Your actual marks

Both use the same scoring rules, so your local number is a genuine estimate - but a detector tuned to the public clips will not carry over, and that gap is the whole reason the private set exists.

The two components

Component	Marks	How it is scored
Lane identification	**20**	Fraction of frames where `lane` matches the ground truth

Component	Marks	How it is scored
Lane centre	**30**	How close `center_x` is to the true lane centre

Lane centre tolerance. A lane is roughly 315 px wide in the frame, so your `center_x` does not have to be exact:

Error against the ground truth	Credit
within **30 px**	full
within **60 px**	half
beyond 60 px	none
the `lane` for that frame was **wrong**	none

That last row matters: the centre of the wrong lane is not the centre of *your* lane, however close the number looks. Get the lane right first.

Frames you give up on still count. Returning `"unknown"` / `-1` on a frame where the lane was clearly visible scores zero on both components, the same as a wrong answer. Reporting failure honestly costs you nothing extra and makes your own debugging far easier

Submission Procedure

1. Install the evaluation tool

Run everything **inside your** `NV_<Team-ID>` **environment** - the one holding Python 3.10, NumPy and OpenCV.

```
conda activate NV_<Team-ID>
pip3 install -U eyantra-autoeval
```

2. Run the evaluation

Run it from `task1c/`, the folder holding your completed `lane_detection.py` with the clips beside it in `public/`:

```
task1c/
├── lane_detection.py
└── public/
    ├── clip_01.mp4
    └── …
```

The `skipped` column is the ambiguous frames - mid lane-change, or the divider out of view - which are excluded for everyone and cost you nothing.

```
eyantra-autoeval evaluate --year 2026 --theme NV --task 1c
```

The tool finds your `lane_detection.py` and the dataset, checks every clip is 640 × 480, then calls your `detect_lane()` on every frame and compares the output against the ground truth. It prints a per-clip breakdown and your public-set score:

```
------------------------------------------------------------------------------
clip              scored  skipped  lane ok  centre ok  mean err
------------------------------------------------------------------------------
clip_01.mp4          200        0      200      200.0      0.2px
...
------------------------------------------------------------------------------
TOTAL               3708      292     3547     3223.0     11.8px
------------------------------------------------------------------------------

  Lane identification :  95.7%  ( 90.2% after the 55.5% chance floor)  -> 18.0
20
  Lane centre         :  86.9%  -> 26.08 / 30
  PUBLIC-SET TOTAL    : 44.13 / 50
```

If it cannot find your script or the clips it will ask for the path. It writes a result file such as `result-2026-NV-1C-20260907.json` into the directory you ran it from.

> ⚠ Run it on the unmodified boilerplate
>
> The tool imports your `detect_lane()` directly. If you renamed the file or the function, changed the keys of the returned dictionary, or left a `cv2.imshow()` / `cv2.waitKey()` inside `detect_lane()`, the run will fail or hang and no result file is written.

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3. Package and upload

Put **two files** into a single zip:

File	What it is
`NV_Task1C.json`	the result file from step 2, renamed
`lane_detection.py`	your completed script

Rename `result-2026-NV-1C-<date>.json` to `NV_Task1C.json`, then zip it together with your script and name the zip `NV_Task1C.zip`.

Ubuntu

```
mv result-2026-NV-1C-*.json NV_Task1C.json
zip NV_Task1C.zip NV_Task1C.json lane_detection.py
```

Windows - rename the result file in Explorer, select both files, right-click and choose **Send to > Compressed (zipped) folder**, then rename the archive to `NV_Task1C.zip`. It does not matter if Windows puts the two files inside a folder within the zip; the grader looks for them either way.

Upload the zip on the eYRC Portal - Task 1.

> ⚠ The filenames are part of the submission
>
> The zip must be named exactly `NV_Task1C.zip`, and your script inside it must still be named `lane_detection.py`.

What the Grader Tells You

After grading, the portal shows a remark. Here is what each one means.

Remark contains	What it means
`Lane identification …%; lane centre …%, mean error …px`	Your score on the private clips, per component
`scored on N of M private frames (K ambiguous frames excluded)`	How many frames counted. Ambiguous frames are excluded for everyone
`N frames returned unknown/-1 where the lane was visible`	Your detector gave up on frames it should have handled; each scored zero

Remark contains	What it means
`Team's own public-set score was X (private Y)`	Your local score next to your real one. A large gap means overfitting to the public clips
`No readable result file in the zip`	The result JSON was missing, renamed wrongly, or edited
`lane_detection.py is not in the submitted zip`	The script was missing or renamed
`Submission called a display function (cv2.imshow / cv2.waitKey) inside detect_lane()`	The grader has no display, and the rules forbid it
`WARNING: the private set is single-lane`	A grader-side note about chance correction
`Exception occurred while evaluating the submission`	Your code raised, or the zip was malformed